

Project Demo



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Problem

In standard SQL, due to the strong coupling between the group by attributes and aggregate functions we are unable to compute multiple aggregates on different subsets of rows in a single query

- Multiple subqueries can result in multiple table calls and multiple joins back to a final relation
- This leads to issues of queries becoming more complex, redundant, and a less efficient and optimized query

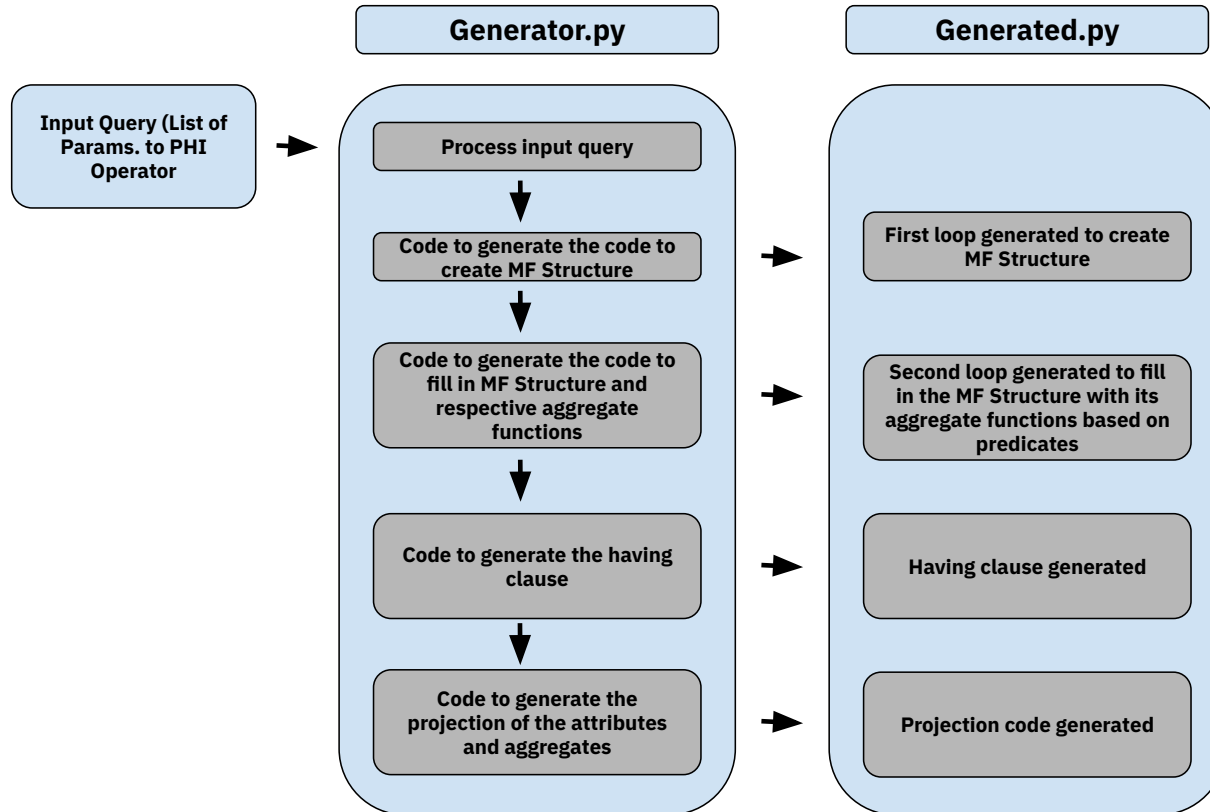
How we address it

- Our project is a query processing engine that runs MF and EMF queries with their extended SQL syntax that includes grouping variables and the such that clause
 - Grouping variable - A variable that references a subgroup of data within the GROUP BY group
 - “such that” clause - Acts as the WHERE clause for the grouping variables
- Technical Significance
 - Allows for multiple aggregates to be computed in a single query
 - Easier to maintain/debug, and allows the sql optimizer to optimize the query

Appendix

1. Architecture of our project
2. Query structure
3. Technologies used
4. Technical Limitations
5. Demo of our project
6. Recap

High Level Architecture

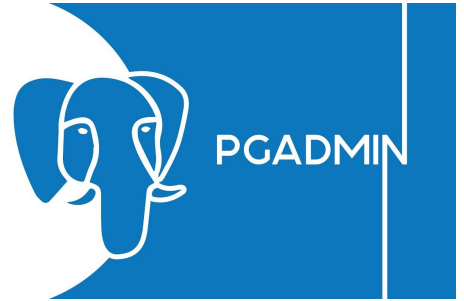
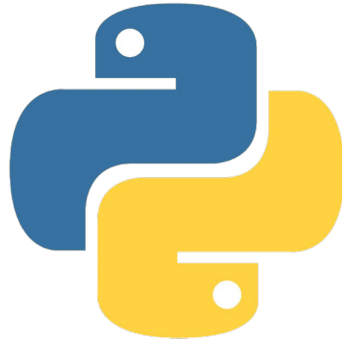


Query Structure

```
SELECT ATTRIBUTE(S):
cust, prod, 1_sum_quant, 2_sum_quant, 3_sum_quant
NUMBER OF GROUPING VARIABLES(n):
3
GROUPING ATTRIBUTES(V):
cust, prod
F-VECT([F]):
1_sum_quant, 2_sum_quant, 3_sum_quant
SELECT CONDITION-VECT([σ]):
1.state='NJ' AND 1.prod = 'Apple'
2.state='NY' AND 2.prod = 'Apple'
3.state='CT' AND 3.prod = 'Apple'
HAVING_CONDITION(G):
1_count_quant > 2_count_quant
```

- Formatted as a list of parameters
- Structured as:
 - <parameter for phi operator>:
 - <argument for the parameter>
- The grouping attributes and grouping variable aggregate functions are separated by commas
- “such that” clause/SELECT CONDITION:
 - Predicate for grouping variable 1
 - Predicate for grouping variable 2
 - and so on
- Having clause is formatted as a string

Technologies



Technical Limitations

- No support for minimal scanning
- Does not support aggregates for entire columns (ex: `1.quant > AVG(quant)`)
- Does not support grouping attributes column in SELECT and HAVING clause (ex: `1.length`)
- No error checking for presence of tables
- Only works for the sales table
- Strict input design

DEMO

Recap and Future Enhancements

Our project aimed to tackle the issue of large, inefficient, and complex queries due to the strong coupling between group by attributes and the aggregate functions by making a query processing engine that handles grouping variables and the “such that” clause

Future Enhancements

- Can be utilized for any table
- More aggregates support
- More flexibility for inputting a query
- Better error handling